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Irish mathematics teachers' attitudes towards inclusion

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This paper through the theoretical framework of constructive attitude theory explores mathematics teachers' attitudes and pedagogical strategies with reference to inclusive practice. The authors argue that though teachers may have formed positive inclusive attitudes, the translation of these into practice does not always occur and poses significant challenges.

Keywords: inclusive education; attitude formation; constructive attitude theory; mathematics teachers

Introduction

Teacher knowledge is a contested area and one that is central to inclusive practice. As mainstream education provides for more diverse learners, teachers' attitudes and expectations are increasingly challenged (McLesky and Waldron 2002). Much of the existing research on teachers' attitudes toward inclusion is based on the experiences of elementary or primary level teachers (DeSimone and Parmar 2006). Less attention is paid to post primary level where there is a stronger academic focus, and where teachers are more likely to be working within subject specific teaching cultures. Ainscow (1997) argues that the best starting point for understanding how classrooms can be made more inclusive is to examine the practices of subject specialist teachers.

This paper through the theoretical framework of constructive attitude theory explores the links between teacher attitudes and pedagogical strategies with reference to inclusive practice. The experience of mathematics teachers is the focus of the study as this is a core area of curricular experience for all students. The authors argue that though teachers may have formed positive inclusive attitudes, the translation of these into classroom teaching is a significant challenge.

Attitude formation

Attitudes are regarded as critical in that they are presumed to guide people's actions (Fishbein and Ajzen 1975). One contemporary theory of attitude formation, which is applicable to inclusive education is constructive attitude theory. This theory advances the view that attitudes are merely evaluative judgments, formed via a constructive process, in the context of decision-making. This theory contends that as people lack the working memory capacity, to hold and retrieve an infinite

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number of well-defined attitudes in memory, they respond according to their contextual goals (Argyriou and Melewar 2011). The model also suggests that attitude foundations are goal specific that require minimum effort in order to achieve accuracy. A significant component of this process is a reliance on heuristic processing strategies in order to arrive at a decision which is not exactly systematic, but not automatic either (Argyriou and Melewar 2011). As a result attitudes are temporary and contextual constructions.

Slinigner, Sherrill and Jankowski (2000) suggest that positive attitudes can influence a positive disposition on the part of teachers teaching students with and without learning difficulties or special educational needs. However, Rose, Shevlin, Winter, and O'Raw (2010) contend that whilst teacher attitudes towards pupils with difficulties may often present as positive, these do not always lead to classroom practices that encourage pupil achievement at the levels, which might be achieved. In order to explore the reasons why this may happen it is necessary to consider the issue of teacher knowledge and practice.

Teachers' knowledge and practice with reference to inclusion

There is an extensive literature concerning teacher knowledge (Ryle 1971; Schulman 1986; Eraut 1994; Fenstermacher 1994; Cochran-Smith and Lytle 1999; Knight 2002). Ryle (1971) referred to teacher knowledge as the distinction between 'knowing that' and 'knowing how'. Eraut (1994) describes four forms of teacher knowledge, propositional knowledge of theories, concepts and pedagogical principles, practical knowledge characterised by routines, procedures, and processes, tacit knowledge, that which is internalised, is known but cannot be told, and 'know-how' the enactment of these in a situation.

The teachers' attitude, and their knowledge is also subject to the contingencies of the social and school context (Shulman 1986). Being a successful teacher of diverse groups of students requires the ability to effectively combine knowledge of the subject content, knowledge of teaching methods, and knowledge about the students (Karp and Voltz 2000). A range of successful inclusive practices, employed by subject specialist teachers in their classrooms has been identified (Florian and Rouse 2001) which, include the use of appropriate differentiation strategies; cooperative learning strategies; classroom management strategies; and teaching social skills. Jordan and Stanovich (1998) found that teachers who were able to include pupils with a wide range of learning abilities did so due their responsiveness to individual need during whole class teaching. Florian and Rouse (2001) concluded that teachers who created inclusive classrooms did not distinguish between students and adopted a problem-solving approach to inclusion. However it also emerged in their study that while teachers reported a high degree of familiarity with inclusive strategies, there was variation between teachers of different subjects in the extent to which they used some of the strategies. These differences between subjects were attributed to a number of factors, which included the nature and status of knowledge in a particular subject domain and whether the teachers perceived learning their subject as being related prior learning. mathematics and modern foreign languages was regarded as sequential while the humanities and English much less so (Florian and Rouse 2001). With reference to mathematics, research suggests that teachers' attitudes about the nature of mathematics shape their personal philosophy about mathematics education and student learning (Cross 2009), which also impacts on their classroom practices.

Mathematics teachers' pedagogical knowledge and inclusive practice

Manouchehri (1997) notes that teachers translate their knowledge of mathematics and pedagogy into practice through the filter of their attitudes. Studies of teachers' attitudes towards mathematics education have investigated teachers' approaches to mathematics teaching (Cobb, Wood, and Yackel 1991). Three philosophies of mathematics teaching have been identified, the problem-solving view, the Platonist view and the instrumentalist view (Thompson 1984). The problem-solving view of mathematics corresponds to the belief that the teacher is a facilitator, the learner is autonomous and learning is the active construction of understanding through problem-solving (Ernest 1989). The Platonist view considers mathematics as a unified body of knowledge where the teacher explains, and the learner receives knowledge with the facility to construct meaning. The instrumental view of mathematics is likely to be associated with a transmission model of teaching, where the strict following of a text or scheme is advocated (Ernest 1989). Each of these views can be influenced by a teacher's attitude about learners' capacity to learn mathematics successfully (Leung 1995). If a teacher, believes the learner does not have the (intellectual) ability or talent to learn mathematics in a problem-solving manner (by actively constructing their own knowledge), the teacher may be tempted to teach mathematics as a set of rules and procedures, adopting an instrumentalist approach (Leung 1995).

Teachers' attitudes, coupled with their knowledge as demonstrated by instructional behaviour are subject to the constraints of school contexts (Van der Sandt 2007), where considerable discussion and debate has focussed on the appropriate setting for learning mathematics. Boaler (1997) found that significant numbers of students in group settings experienced difficulties working at the pace of the class, and became disillusioned and de-motivated by the limits placed upon their mathematical achievement within their setting. Boaler, Dylan, and Brown (2000) concluded that ability grouping was associated with curriculum polarisation that did not serve students in lower settings well. Equally for students in higher settings they were forced to learn at a pace, which was, for many incompatible with understanding.

With the diversity of ability levels in inclusive classrooms, a variety of teacher-centred to student-centred learning approaches in mathematics is beneficial (Karp and Voltz 2000). Self-instruction and direct instruction are important teaching approaches for students with special needs (Kroesbergen and Van Luit 2003). Fuchs and Fuchs (2005) found that combing self-regulated learning strategies with explicit instruction and peer tutoring led to positive outcomes for students (Fuchs and Fuchs 2005). The use of manipulative devices and visual images can help the process of problem-solving and visual–spatial representations improve students' capacities to solve problems (Van Garderen 2006). The use of technology has been found to enhance learning by providing opportunities for exploration and concept representation (Gagnon and Maccini 2001). Both inclusive education and mathematics education have undergone significant debate and discussion within the Irish education system.

Inclusion – the Republic of Ireland context

The Irish education system is highly centralised and organised by the Department of Education and Skills. At second level there are secondary (384) vocational, (254) comprehensive schools (14) and community schools (78) (Department of Education

and Skills 2011). The majority of pupils transfer to second level school when they have completed the full first level course – generally at about 12 years of age (Department of Education and Science 2004).

In common with other countries a number of significant policies focused on inclusion have been implemented in the Republic of Ireland in recent years (Meijer 2003). Rose et al. (2010) point out that within recent decades there were a series of significant government-sponsored reports highlighting inadequacies in special education provision (Report of the Special Education Review Committee 1993; Commission on the Status of People with Disabilities 1996), parental litigation challenging existing provision (Supreme Court of Ireland 1996, 2001) and groundbreaking legislation guaranteeing curricular access for children and young people with special educational needs (Education Act 1998, Education for Persons with Special Educational Needs [EPSEN] 2004). The Disability Act 2005, supporting the aims of the EPSEN Act 2004, further strengthened the rights of children with disabilities. However the EPSEN Act has not yet been fully implemented.

Research on inclusion within an Irish context has focused on the areas of policy, provision, experience and outcomes (Rose et al. 2010). Individual studies (O'Brien 1992; Kennedy 1994; Shevlin and O'Moore 1999) have produced some useful insights into inclusive practice. The recent initiatives in inclusive education have changed the nature of Irish classrooms but few studies have examined the impact of these changes on mainstream subject specialist teachers. Mathematics education as an area has not been researched extensively.

Close and Oldham (2005) argue that the examination and assessment tradition in Ireland promotes a certain type of mathematical competency and draws upon narrow range of mathematical situations. The introduction of Project Maths (2010) into the second level curriculum is a major programme of reform of mathematics and is designed to ensure an appropriate balance between the understanding of mathematical theory and concepts and the development of practical applications skills. It will take some time to establish whether this new approach has been successful.

To date due to enormous pressure to achieve high grades mathematics teachers generally teach to the examination, training students how to pass tests, and adopt teaching styles which in many cases do not match the preferred learning style of many students (NCCA 2005). Over the eight-year period of 2001–2008, the failure rate for ordinary and foundation levels mathematics in the Leaving Certificate examination at post-primary level was 13% and 7% respectively. The percentage of pupils achieving E, F and NG remained reasonably stable across the eight-year timeframe (Doyle 2010). This could be attributed to the fact that many students failing at ordinary and foundation levels may not have received adequate teaching and supports in mathematics and could also suggest that these students have undiagnosed learning difficulties or specific learning problems in mathematics (Doyle 2010). These issues suggest the need for a closer examination of mathematics teachers' preferred classroom practices and their attitudes towards inclusion.

Methodology

This study employed a mixed survey instrument which contained both closed-ended questions and projective sentence completion (Appendix). This is labeled as intra-method mixing (Johnson and Turner 2003) and facilitates within method triangulation (Denzin 1970). Combining closed-ended and projective sentence completion has the

potential to capture the benefits of both quantitative and qualitative data collection and analysis (Erickson and Kaplan 2000). Projective sentence completion was used to provide respondents with the opportunity to express opinions and feelings that would have been difficult to access by direct questioning (Oppenheim 1992). The appeal of this approach is that it provides a rich description of respondent reality and can also capture diversity in responses and provide alternative explanations to those that closed-ended survey questions are able to capture (Pothas, Andries, and DeWet 2001).

The key to determining the status of data derived from projective sentence completion is their depth; both the amount recipients are prompted to write and the amount they actually write (O Cathain and Thomas, 2004). The respondents were presented with the start of a sentence (Calterall and Ibbotson 2000), 'For me inclusion means…'. At the design stage of the study the researchers decided to generate depth and treat the data qualitatively. This was achieved through the provision of instructions on the questionnaire and additional space for respondents to offer their perspectives about inclusion. The quality of the responses to the projective sentence completion raised a number of issues not explored in the closed questions and merited further analysis.

Self-report methodologies have been criticised due to the issue of social desirability and response bias (Razavi 2001). Socially desirability refers to the tendency for people to give replies that make them appear open minded and unprejudiced. Ajzen (1996) contends that simple questions about people's attitudes and intentions have considerable predictive validity, in that they often correlate highly with people's overt behaviour. Fisher (1993) found that indirect questioning reduced social desirability bias on variables subject to social influences and this provided the rationale for the inclusion of the projective sentence completion element.

Data collection and analysis

Data were generated via a self-completion questionnaire. The instrument sought to collect:

- Biographical and demographic data about the teachers.
- Their preferred approaches to teaching mathematics.
- Their attitudes towards inclusion.

The instrument was piloted with three practicing mathematics teachers and was completed without any difficulties. Respondent driven sampling or snowball sampling, a non-probability method of sampling was used to access mathematics teachers. The researchers chose this method to encourage maximum response rates across a number of different school types as gaining, access to the full sample of mathematics teachers in the Republic of Ireland was not possible. Contact was made with teachers in different types of post-primary schools across the country. These contacts agreed to distribute the questionnaires to mathematics teachers in their schools.

In total 160 questionnaires were distributed with 121 returned completed, giving a response rate of 76%. The sample that was returned yielded a good cross-section of teachers working in different type of schools. The majority of responses came from voluntary secondary schools, which are the largest number in the Irish education system, followed by vocational schools, which have the next highest number.

The remaining respondents were distributed evenly between community/comprehensive schools and private schools, which represent the least number of schools in the system (see Table 2). This resulted in a good cross section of mathematics teachers and facilitated a number of different perspectives. Ethical approval was received from the university in which the study was conducted. Consent to participate was indicated on the survey instrument which allowed each respondent to indicate their agreement as participants in the study. The study does not seek to generalise the findings in relation to mathematical and inclusive education in the Republic of Ireland, which is a limitation, nevertheless some important themes emerge from the findings that merit consideration and reflection for future studies in this area.

Quantitative data: survey questionnaire

Data obtained were analysed using the Statistical Package for the Social Sciences (SPSS version 15.0). Descriptive statistics were used to summarise demographic data and the results from the questionnaire.

Qualitative data: survey questionnaire

The content analysis in this study identified a comment as the minimum size of a recordable unit. Krippendorf (2004) described recordable units as 'units that are distinguished for separate description and coding' (100). The responses to the projective sentence completion were coded and content analysed using guidelines for analysing qualitative data (Miles and Huberman 1994). The primary researchers identified major themes, developed an initial set of themes, and examined the theme categories for consistency in meaning and context. These were iteratively refined using the constant comparison method (Krippendorf 2004). The categorisation of comments by theme was conducted separately and blindly, by the primary researchers.

The findings of the statistical analyses are outlined, which presents the demographic profile of the respondents, their preferred approaches to teaching mathematics and their views about inclusion. The qualitative analyses are presented which reports their attitudes and classroom practices with reference to inclusion. These findings are discussed using constructive attitude theory as a conceptual lens with reference to relevant research in the area. The implications for inclusive education are then presented.

Results

The age, gender and degree profiles of the cohort and patterns of association between these variables

The age group and gender profiles of the teachers who were surveyed are outlined in Table 1.

The majority of respondents were over 40 years and were female. The gender balance of the sample was similar to the trend internationally (Drudy et al. 2005). Over 40% of the respondents had been teaching for less than 10 years. Data concerning school classification is outlined in Tables 2.

The majority of respondents worked in voluntary secondary schools (60%). All teachers in the sample had experience working in inclusive classrooms. Dyslexia and mild learning disabilities pre-dominated (65%) and ADD (28%) ranked highest in relation to emotional and behavioural difficulties.

Table 1. Gender and age profiles of respondents (n = 121).

	% of respondents	No of respondents	Age range of respondents	% of age group	No of respondents
Males	31%	(n = 38)	Under 25 yrs	7%	(n = 8)
			25 but under 30 yrs	21%	(n = 26)
Females	69%	(n = 83)	30 but under 40 yrs	29%	(n = 35)
			40 yrs and over	43%	(n = 52)

Table 2. Classification of school in which respondents worked (n = 119).

Type of school	Total %	Gender	%
Voluntary secondary school	60%	Males	57%
	(n = 71)	Females	61%
Vocational school/community college	14%	Males	11%
	(n = 17)	Females	16%
Private school	13%	Males	11%
	(n = 15)	Females	13%
Community/comprehensive school	13%	Males	22%
	(n = 16)	Females	10%

Preferred approaches to teaching mathematics

Patterns of respondents' preferred approaches to teaching mathematics on a Likert scale rated strongly agree to strongly disagree are presented in Table 3.

A number of statements elicited high levels of agreement, these included: 'Students need to proceed in a logical structure when learning mathematical concepts' yielded agreement of the majority of respondents (96.7%), followed by 'I use class tests to establish students knowledge' (95.8%) and 'I encourage students to use relevant mathematical terms when appropriate' (89.2%) and 'I explain and teach the language of mathematics' (89.2%). Statements which indicated a more problem-solving approach to teaching the subject and which reflected practices that are considered helpful to students' learning, such as using group discussion (55.3%), pair work (54.5%), ICT (33.1%) and using fun games (32.3%), elicited lower levels of agreement with reference to their preferred approaches to teaching mathematics.

Attitudes towards the inclusion of students with special educational needs in mainstream classes

Respondents' levels of agreement with reference to the inclusion of students with special educational needs in mainstream classes are outlined in Table 4.

A majority of respondents (90%) agreed that inclusion promoted understanding and acceptance of differences. Eighty-nine per cent of respondents, agreed with the view that students with SEN should be provided with opportunities to participate in a meaningful way in the full range of activities in the classroom. Eighty per cent of respondents agreed that lack of time and pressure to cover the course was the main barrier to successful inclusive practice. Over 70% of respondents agreed that mainstream schooling did not benefit all students. With reference to teaching approaches 50% of respondents indicated that they differentiated modes of assessment (including homework) in relation to students' ability. Fifty two per cent believed that more

Table 3. Preferred approaches to teaching mathematics.

Rank	Statement	Item scores Mean	Item scores SD	Agreement %
1	Students need to proceed in a logical structure when learning mathematical concepts	1.54	0.620	96.7%
2	I use class tests to establish students knowledge	1.48	0.579	95.8%
3	I encourage students to use relevant mathematical terms when appropriate	1.80	0.641	89.2%
4	I explain and teach the language of mathematics	1.74	0.665	89.2%
5	It is possible to use a variety of strategies and approaches in teaching maths	1.84	0.683	88.5%
6	I revisit and review previously learned knowledge regularly	1.85	0.760	86.0%
7	I keep periods of instruction short and to the point	2.02	0.796	80.2%
8	I vary activities to help students learn concepts	2.15	0.803	76.8%
9	I use the text book as my main resource in my maths teaching	2.08	1.005	76.1%
10	I use visual clues to aid memory	2.15	0.715	74.4%
11	My board work is always carefully presented	2.09	0.856	71.9%
12	I practise estimation skills so students can use the calculator efficiently	2.23	0.834	70.2%
13	I outline learning objectives at the beginning of every lesson	2.21	0.868	66.2%
14	I review what has been learned at the end of every lesson	2.24	0.857	63.7%
15	I get students to repeat given instructions	2.45	0.974	61.1%
16	I encourage group discussion when I introduce a new theme or concept	2.54	1.033	55.3%
17	I actively encourage pair work and group work in class	2.60	1.077	54.5%
18	I have clear rewards for staying on task and completing it	2.64	0.865	43.8%
19	I use ICT to help students learn concepts	2.98	1.076	33.1%
20	I use fun games such as table-darts or fraction–decimal equivalence dominoes to help the automatisation of certain operations	3.15	1.078	32.3%

able students may be held back due to the presence of students with SEN in the classroom as the teacher cannot teach at the pace that they need to be challenged. Half of the respondents agreed with the view that students with SEN in mathematics class did not receive the attention that they deserved. Just under half of the respondents (48.7%) agreed that withdrawal from the mainstream class for individual or small group work was better.

The questionnaire through projective sentence completion allowed respondents to give their own personal perspectives with reference to inclusion. Key themes that emerged from the analysis related to the teachers' attitudes and their classroom practices with reference to inclusion. Verbatim comments are used to illustrate themes and are referenced according to the question and page numbers of the transcripts.

Teachers' attitudes to inclusion

Respect was considered important in meeting the needs of all students in the classroom in a positive learning environment. This is reflected in one respondent's comment: 'respecting and accommodating the needs and abilities of all students in your

Table 4. Attitudes toward inclusion of students with SEN in mainstream classes.

Rank	Statement	Item scores Mean	Item scores SD	Agreement %
1	Inclusion promotes understanding and acceptance of differences	1.74	0.704	90.1%
2	Students with SEN should be provided with opportunities to participate in a meaningful way in the full range of activities in the classroom	1.73	0.645	89.3%
3	The lack of time and pressure to cover the course is the main barrier to successful inclusive practice	1.67	1.037	80.1%
4	There is a lack of training for mainstream teachers teaching students with SEN	1.80	0.954	78.5%
5	I change my teaching approaches to accommodate students with SEN	1.98	0.806	76.8%
6	Differentiation refers to the ways that teachers take into account the differences among their students	1.92	0.770	76.0%
7	Mainstream schooling does not benefit all students	2.02	1.049	71.1%
8	I am fully aware of my role and responsibilities regarding students with SEN	2.07	1.034	66.9%
9	Inclusion is very beneficial to all students in the class	2.03	1.080	62.0%
10	I differentiate modes of assessment (including homework) in relation to students' ability	2.21	1.048	57.0%
11	I feel confident in my ability to teach maths successfully to students with SEN	2.39	1.165	55.4%
12	More able students may be held back as you cannot teach at the pace they need to be challenged	2.37	1.219	52.1%
13	Students with SEN in maths class do not receive the attention they deserve	2.48	1.219	50.4%
14	Withdrawal from the mainstream class for individual or small group is better	2.40	1.045	48.7%
15	It is possible to differentiate instruction to reach every student in the class to help them realise their individual potential	2.57	1.161	48.7%
16	Differentiation is grouping students according to ability	2.69	1.189	42.1%
17	There is more distraction/disruption in class	2.69	1.154	34.7%
18	Mainstream schooling can have a negative affect on students with SEN	2.74	1.109	33.9%
19	Inclusion of students with SEN in the mainstream classroom holds back the academic achievement of the rest of the class	2.82	1.190	33.1%
20	Special classes in the mainstream school is better for students with SEN	2.79	1.125	28.9%
21	Other students do not accept the difficulties experienced by those with SEN in maths	3.04	1.121	24.8%
22	Students with SEN are the responsibility of the learning support team	3.47	1.245	15.7%

classroom where all students learn together in a positive environment with respect' (CQ6; 4). It was also considered important that students should not be made feel inferior. Respondents' opinions were characterised by such remarks as: '[students should be] part of mainstream and not made feel inferior but with extra help given to support the main class' (CQ6; 4). 'Allowing students to learn with their peers without stigmatising them' (CQ6; 6). Ensuring that students feel valued also emerged in the responses. One respondent defined inclusion as: 'all students are

taught together in school with each pupil valued and no student left behind' (CQ6; 7). Accepting and recognising difference emerged as key elements in their attitudes towards inclusion. Many respondents agreed with the view that: 'recognising all individuals abilities potential strength and weakness and providing opportunity for all to reach their maximum potential, accepting difference' (CQ6; 8). One respondent commented:

> … that students with disabilities are not treated any differently but as their teacher it is important to be aware of their disability and tend to their needs on an individual basis without drawing too much attention to them at the same time. (CQ6; 9)

Equality of opportunity featured in the responses typically characterised by the following comment: 'providing all students with the same opportunities in school and helping those who are experiencing difficulties through learning support, etc' (CQ6; 11). One respondent commented: 'every student whatever their ability or inability, are taught to the best of a teacher's ability and no favouritism should be shown to more talented and able students' (CQ6; 12). The social aspects of inclusion also featured in the responses: 'giving students ownership of the social and educational processes in school' (CQ6; 25). 'It is very much a journey never a destination' (CQ6; 26). In many of the responses there was a focus on student being happy in class. This view was characterised by the following comment: 'making all students feel happy and welcome in the class; helps students become aware of difference that people have' (CQ6; 18). Ensuring that students felt comfortable in class was a generally held view as reflected in the remarks of one respondent who stated: 'being part of the class group in as far as one is comfortable' (CQ6; 19). For some mathematics teachers developing a sense of self worth was considered important: 'to feel that one is a worthwhile member of a group and being with everybody else' (CQ6; 20).

Teachers' classroom practices with reference to inclusion

For many respondents encouraging student participation was central to inclusive practice in the classroom. Facilitating students' enjoyment of participating in class, taking part in class activities and being part of a larger group featured in their responses: 'Everyone enjoys the class, and does not feel left out or stupid during class' (CQ6; 16). 'Taking part in everyday class activities, making all students feel that they are part of a larger group' (CQ6; 17). One respondent commented: 'that every student in my class feels part of the class and is involved in classroom activities in some way' (CQ6; 18). Another respondent explained: 'all students working together on a common topic' (CQ6; 18).

Motivating students to participate and developing their sense of achievement was considered important. This was encapsulated in the following statement: 'trying to the best of my ability to encourage, motivate and include all students in my classes to achieve to the best of their ability' (CQ6; 21). Being a full participant in class and developing a sense of achievement was also considered important: 'the student feels that they are part of the class and that they can get a sense of achievement everyday' (CQ6; 22). There was also a focus on incremental achievement: 'meeting the needs of the students whether in a mainstream class or in a special class working at a level' (CQ6; 23). 'That they can cope with and that they feel they can achieve something however small a step it may be' (CQ6; 24).

Building students' confidence was also considered important:

> … getting the student to realise that they can cope with the subject, building confidence is the first step to inclusion, the second step is to find the correct level so the student can maximise their ability. (CQ6; 27)

The development of potential also featured in responses: 'creating a learning environment where all students can achieve their full potential' (CQ6; 28). 'Making sure all the students in the class reach full potential' (CQ6; 29). One respondent commented:

> … that students with disabilities are not treated any differently but as their teacher it is important to be aware of their disability and tend to their needs on an individual basis without drawing too much attention to them at the same time. (CQ6; 9)

Many respondents focussed on the delivery of lessons where the needs of all students are addressed: 'teaching every student who sits in front of me regardless of their ability i.e., adopting strategies to suit individuals so that everyone feels part of the class regardless of ability' (CQ6; 31). Some respondents focused on pace and content suggesting that: 'a meaningful learning experience for the student [is created] where pace and content are both within the grasp of the student of any ability who applies him/herself' (CQ6; 32). The development of skills in mathematics as an aspect of inclusion was also considered important: 'giving all our students an opportunity to develop maths skills and develop confidence in general' (CQ6; 35).

Discussion

The findings from this study from a constructive attitude theory perspective suggest that understanding the attitudes and practices of mathematics teachers with reference to inclusion is an important area of study. Central to the model are factors such as context, goal orientation, and heuristic processing strategies. The model provides a useful lens through which attitudes and practices can be viewed. While no generalisations can be made from the sample used nevertheless the data provides some interesting insights.

Mathematics teachers in this study had experience working in inclusive classrooms and were positively disposed towards inclusion. With reference to Ryle's (1980) concept of teacher knowledge it is clear that mathematics teachers in this study knew the 'that' of inclusion, they focussed on the importance of respect, valuing students, promoting equality of opportunity, ensuring that students were happy and comfortable in class and promoting their students sense of self worth. With reference to the 'how' of inclusion they focussed on class participation, motivation, the importance of incremental achievement, teaching at appropriate levels and responding to individual needs. Yet this differed from their preferred teaching strategies in achieving their goals in mathematics as a subject.

Understanding goal orientation is an important element of constructive attitude theory. For mathematics teachers in this study the 'how' of teaching mathematics was predominantly instrumentalist (Thompson 1984; Ernest 1989) with an emphasis on a transmission model of teaching, following logical structures, using class tests and explaining the language and terminology of mathematics. While they referred to inclusive practices in their classrooms their preferred teaching approaches did not

prioritise group and pair work; using ICT or games in class. They did not employ a variety of teacher-centred to student-centred learning approaches, which are necessary to promote student learning in inclusive mathematics classrooms (Boaler 1997; Karp and Voltz 2000; Kroesbergen and Van Luit 2003; Fuchs and Fuchs 2005; Van Garderen 2006; Gagnon and Maccini 2001).

In constructive attitude theory the affect of heuristic processing strategies is significant and this also applies to mathematics teachers in this study. They know from experience of working within inclusive classrooms the importance of creating a sense of belonging, comfort and happiness for all students, yet few provided concrete pedagogical strategies that facilitated inclusive practice, a finding that concurs with the views expressed by Rose et al. (2010). This has a number of implications for inclusive education.

Implications for inclusive education

The preceding analysis has important implications for mathematics teachers working in inclusive classrooms. Constructive attitude theory assumes that attitude formation is the result of the interaction between contextual variables, goal orientation and heuristic processing strategies. Hence the resultant attitude formations are temporary and contextual constructions. This reality points to the need for more focussed research on the ways in which mathematics teachers form their attitudes towards inclusion and how they translate these attitudes during the practice of their teaching. Equally important is the need to facilitate mathematics teachers' thinking about their preferred teaching strategies within the classroom context to ascertain their inclusivity.

The findings from this study situated within constructive attitude theory suggests that support for mathematics teachers working in inclusive education contexts should allow them to thoroughly examine their own attitudes in the context of their teaching strategies, their school contexts and the wider discourses that inform the area. Such an approach would allow a process of reflection with reference to inclusive education that is often absent. The study highlights the importance of constructive attitude theory as a useful tool in exploring attitudes and resultant teaching practices. This article argues that it is only through such analysis that insights can be gained about the needs of mathematics teachers in their classrooms. The implications of this, for the promotion of inclusive practice, are of fundamental importance to all students in the education system.

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Appendix. Teaching Maths in Inclusive Classrooms Questionnaire

Dear Teacher

The following questionnaire measures your attitudes to teaching maths in classes where students with special educational needs are present. We would be very grateful if you would complete the survey. Participation is voluntary. Your answers are completely confidential and will be reported so that no individual's answers can be identified.

Please tick the following box if you agree to complete the questionnaire □

Thank you for your help

Elaine Whitty and Marie Clarke

Section A

The following background information is being requested solely to enable me to profile maths teachers across these categories

[1] Please indicate your gender. *Tick (✓) the box that applies.*
Male □ Female □

[2] Please indicate in which of the following age groups you are. *Tick (✓) the box that applies.*
Under 25 □ 26–30 □ 31–40 □ Over 40 □

[3] Please indicate how many years you are teaching. *Tick (✓) the appropriate box.*
0–10 yrs □ 11–20 yrs □ 21–30 yrs □ 30+ yrs □

[4] What type of school do you work in? *Tick (✓) the appropriate box.*
Voluntary secondary school □ Vocational school/community college □
Private □ Community/comprehensive school □

[5] What gender do you teach? *Tick (✓) the appropriate box.*
All girls □ All boys □ Mixed □

[6] Please enter the type of primary degree you hold in the space provided (e.g., BA, BComm).
_____

[7] Please indicate the subject area that is the major component of your qualification?
Tick (✓) the box that applies.

Maths	□	Applied maths	□	Statistics	□
Maths physics	□	Chemistry	□	Applied chemistry	□
Physics	□	Applied physics	□	Biology	□
Geography	□	Agricultural science	□	Computers/IT	□
Business	□	Economics	□	Accountancy	□
Engineering	□	Mechanics	□	Technical drawing	□

Arts/humanities (other than mathematics) □
Other subjects (please specify) _____

[8] What is the highest qualification that you have obtained? (e.g., Degree, H.Dip., Masters)
_____

[9] Please indicate the type of third level institution attended. *Tick (✓) as many boxes as are applicable.*
University □ Primary teacher training college □
Institute of technology □ Institution in another jurisdiction □
Other (please specify) _____

[10] Please indicate at what level you teach maths during this year. *Tick (✓) all that apply.*

Junior Certificate:	Foundation	□	Ordinary	□	Higher	□
Leaving Certificate Established:	Foundation	□	Ordinary	□	Higher	□
Leaving Certificate Applied:	Common	□				

Section B

The following statements are designed to identify your attitudes about your experience of teaching maths.

Each item has 5 possible responses. The responses range from '1' (*strongly agree*) through '2' (*agree*) to '3' (*neither agree nor disagree*), '4' (*disagree*) to '5' *(strongly disagree*).

Please read each statement. Mark the *one* response that most clearly represents your degree of agreement or disagreement with that statement. Please respond to all of the statements.

		Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree
1.	I concentrate my teaching in developing student capacity with maths rather than focusing on the subject	1	2	3	4	5
2.	Students should be able to apply their knowledge of maths to real life situations	1	2	3	4	5
3.	The maths syllabus at Junior Certificate foundation and ordinary level does not stretch students sufficiently in the subject area	1	2	3	4	5
4.	The higher level maths course at Junior Certificate is too broad	1	2	3	4	5
5.	There is too large a gap between ordinary and higher level maths at senior cycle	1	2	3	4	5
6.	Many students have a negative view of maths due to their experiences prior to coming to secondary school	1	2	3	4	5
7.	I use active teaching methodologies in all my maths classes	1	2	3	4	5
8.	I ensure that I prepare different questions of varying standards so that all students in my class can have a sense of achievement in the subject	1	2	3	4	5
9.	It is possible to use a variety of strategies and approaches in teaching maths	1	2	3	4	5
10.	Students need to proceed in a logical structure when learning mathematical concepts	1	2	3	4	5

(*Continued*)

Appendix. (*Continued*).

		Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree
11.	I use visual clues to aid memory	1	2	3	4	5
12.	I practise estimation skills so students can use the calculator efficiently	1	2	3	4	5
13.	I use fun games such as table-darts or fraction–decimal equivalence dominoes to help the automatisation of certain operations	1	2	3	4	5
14.	I keep tasks short to accommodate students who have short attention span or poor concentration	1	2	3	4	5
15.	I have clear rewards for staying on task and completing it	1	2	3	4	5
16.	I keep periods of instruction short and to the point	1	2	3	4	5
17.	I observe and use preferred learning styles in planning future work	1	2	3	4	5
18.	I encourage group discussion when I introduce a new theme or concept	1	2	3	4	5
19.	I vary activities to help students learn concepts	1	2	3	4	5
20.	I use ICT to help students learn concepts	1	2	3	4	5
21.	I contextualise content and use real-life problems relevant to the student's experience	1	2	3	4	5
22.	I actively encourage pair work and group work in class	1	2	3	4	5
23.	I get students to repeat given instructions	1	2	3	4	5
24.	When I am discussing area or volume I use a model of the relevant shape	1	2	3	4	5
25.	I revisit and review previously learned knowledge regularly	1	2	3	4	5
26.	I use cross-curricular approach to the teaching of skills or concepts that are common to different subjects, e.g., measuring angles in technical graphics and maths; PAYE calculations in maths and business studies	1	2	3	4	5
27.	I encourage students to make connections to mathematical concepts that may be encountered in other areas of the curriculum, e.g., *'How did you find the volume of something in science?'*	1	2	3	4	5

(*Continued*)

Appendix. (*Continued*).

	Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree
28. My board work is always carefully presented	1	2	3	4	5
29. I encourage students to use relevant mathematical terms when appropriate	1	2	3	4	5
30. I explain and teach the language of mathematics	1	2	3	4	5
31. I use class tests to establish students knowledge	1	2	3	4	5
32. I use the text book as my main resource in my maths teaching	1	2	3	4	5
33. I outline learning objectives at the beginning of every lesson	1	2	3	4	5
34. I review what has been learned at the end of every lesson	1	2	3	4	5

Section C

Post-primary schools now have much more diverse student populations than was the case in the past.

I wish to ascertain if you have young people in your classes with learning disabilities or emotional and behavioural disorders and what supports are available (Q3)

[1] Do you have any student(s) in the classes you have been assigned to teach this year who have LEARNING DISABILITIES?

YES □ NO □

If YES, how many such students in total do you have in all your classes?

_____

[2] Do you have any student(s) in the classes you have been assigned to teach this year who have EMOTIONAL AND BEHAVIOURAL DISORDERS?

YES □ NO □

If YES, how many such students in total do you have in all your classes?

_____

[3] If you answered YES to Q1 or Q2 please indicate which of the disabilities listed below these student(s) have been assessed as having/you have been informed about. *Tick (✓) as many boxes as are applicable*.

Learning disabilities

Dyslexia/specific learning disabilities □
Autism/Asperger syndrome □
Mild general learning disabilities □
Down syndrome □
Speech and language disorders □

Emotional and behavioural difficulties

Attention Deficit Disorder (ADD) □
Attention Deficit and Hyperactivity Disorder (ADHD) □
Tourette syndrome □

Physical disabilities – sensory impairments
Hearing impairment ☐
Visual impairment ☐
Mobility impairment ☐

Other (please specify) _____

[4] If you answered YES to Q1 or Q2, are you being offered any help or support in the school in helping you to meet these students' needs?
YES ☐ NO ☐
If YES, please describe briefly the type of help or support you are being given.

[5] What is the school policy with regards to including students with Special Educational Needs (SEN)? *Tick (✓) as many boxes as are applicable.*
Mainstream class, full time with withdrawal for individual instruction ☐
Mainstream class, with withdrawal for group instruction ☐
Mainstream class, supported by in-class teaching, with some withdrawal ☐
Mainstream class, supported by team teaching with no withdrawal ☐
Mainstream class, supported by Classroom Assistants ☐
Mainstream class with occasional advice from Learning Support ☐
Mainstream class part-time and special class part-time ☐
Mainstream class with no additional support ☐
Special class full-time ☐
Team-teaching ☐
Don't know ☐
Other (please specify) _____

[6] Please complete this sentence. Use additional pages provided.
For me, inclusion means…

[7] What is your opinion of the inclusion of students with special educational needs (SEN) in the mainstream class?

		Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree
1.	Inclusion is very beneficial to all students in the class	1	2	3	4	5
2.	Mainstream schooling does not benefit all students	1	2	3	4	5
3.	Mainstream schooling can have a negative affect on students with SEN	1	2	3	4	5
4.	Inclusion promotes understanding and acceptance of differences	1	2	3	4	5
5.	Inclusion of students with SEN in the mainstream classroom holds back the academic achievement of the rest of the class	1	2	3	4	5
6.	Withdrawal from the mainstream class for individual or small group is better	1	2	3	4	5
7.	Special classes in the mainstream school is better for students with SEN	1	2	3	4	5

(*Continued*)

Appendix. (*Continued*).

		Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree
8.	Students with SEN are the responsibility of the learning support team	1	2	3	4	5
9.	More able students may be held back as you cannot teach at the pace they need to be challenged	1	2	3	4	5
10.	There is more distraction/ disruption in class	1	2	3	4	5
11.	Other students do not accept the difficulties experienced by those with SEN in maths	1	2	3	4	5
12.	There is a lack of training for mainstream teachers teaching students with SEN	1	2	3	4	5
13.	Students with SEN in maths class do not receive the attention they deserve	1	2	3	4	5
14.	I feel confident in my ability to teach maths successfully to students with SEN	1	2	3	4	5
15.	I change my teaching approaches to accommodate students with SEN	1	2	3	4	5
16.	Students with SEN should be provided with opportunities to participate in a meaningful way in the full range of activities in the classroom	1	2	3	4	5
17.	I am fully aware of my role and responsibilities regarding students with SEN	1	2	3	4	5
18.	I differentiate modes of assessment (including homework) in relation to students' ability	1	2	3	4	5
19.	Differentiation is grouping students according to ability	1	2	3	4	5
20.	Differentiation refers to the ways that teachers take into account the differences among their students	1	2	3	4	5
21.	It is possible to differentiate instruction to reach every student in the class to help them realise their individual potential	1	2	3	4	5
22.	The lack of time and pressure to cover the course is the main barrier to successful inclusive practice	1	2	3	4	5

THANK YOU FOR YOUR COOPERATION. IT IS MUCH APPRECIATED.